DIKSHIT DHAKAL

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PROFILE

Final-year Civil Engineering student at IOE Thapathali Campus specializing in machine learning applications to hydrologic modeling, streamflow regionalization, and climate change sensitivity for hydropower engineering. Proven track record in developing programmatic QA/QC pipelines, data-driven streamflow prediction models, and spatial analytics using Python, XGBoost, and QGIS. Seeking a Graduate Research Assistantship (MS) in Water Resources Engineering.

EDUCATION

Bachelor of Engineering in Civil Engineering **(Expected Graduation: July 2027)**

Institute of Engineering (IOE), Thapathali Campus, Tribhuvan University, Nepal

- **Relevant Coursework:** Engineering Hydrology, Hydraulics, Fluid Mechanics, Hydropower Engineering, Computational Techniques in Civil Engineering, Numerical Methods, Probability & Statistics, Irrigation & Drainage.

RESEARCH EXPERIENCE

Undergraduate Thesis / Final Year Research (In Progress) **(2026 – Present)**

Machine Learning-Based Streamflow Regionalization & Multi-Model Comparative Framework & Climate Sensitivity for Hydropower Potential (Daraudi River Basin)

- **Multi-Model Benchmark Pipeline:** Formulated a comprehensive hydroinformatics pipeline comparing 4 benchmark methods Structural Time Series (STS), Self-Organizing Maps (SOM), XGBoost, and Random Forest to evaluate prediction accuracy in data-scarce catchments.
- **26-Station Calibration & Data Filling:** Executed automated time-series calibration, quality control, and missing data gap-filling across 26 hydrological gauge stations in the region to construct a robust regional streamflow dataset
- **Prediction in Ungauged Basins (PUB):** Extracted spatial catchment descriptors (DEM, slope, LULC, soil properties via ISRIC SOTER) using QGIS and WhiteboxTools in Python, building a z-score normalized physical similarity index to select donor basins..
- **Climate Sensitivity & Hydropower Optimization:** Coupled regionalized streamflow with downscaled CMIP6 far-future climate scenarios to quantify non-stationary shifts in Flow Duration Curves (FDC), dry-season water deficits, and design discharge sensitivity (Q_{40} to Q_{95}).

PROJECTS AND SOFTWARE DEVELOPMENT

Discharge Gap-Filling & QA/QC Automated Pipeline (Gandaki Basin)

- **Pipeline Engineering:** Built an automated machine learning and time-series pipeline in Python to reconstruct missing daily discharge records across 26 stations using physical donor relationships.
- **Hydraulic Quality Control:** Integrated upstream-downstream hydraulic consistency checks and outlier detection algorithms, iteratively retraining models until strict Percent Bias (PBIAS) targets were satisfied across 26 stations and adjacent basins.
- **Model Integration:** Designed and implemented a modular Python application integrating Linear Regression, Polynomial Regression, Random Forest, and Gradient Boosting models for discharge forecasting.

Structural Dynamics & Experimental Mechanics

- **Shake Table Testing:** Co-designed and manufactured a functional dual-axis two-directional shake table for dynamic testing of structural scale models.

TECHNICAL SKILLS

- **Hydroinformatics & Spatial Tools:** QGIS, WhiteboxTools, HEC-HMS, ETABS, AutoCAD
- **Programming & Machine Learning:** Python (Pandas, NumPy, Scikit-learn, XGBoost, PyTorch/TensorFlow for LSTM)
- **Data Analytics:** Time-series analysis, QA/QC automation, CMIP6 downscaling, spatial catchment descriptors
- **Tools & Environments:** VS Code, Git

LEADERSHIP AND SERVICES

Vice President | Civil Engineering Student Society (IOE Thapathali Campus)

- **Leadership:** Directed institutional engineering events and organized an inter-college structural modeling competition.

Special Guest Lecturer | Department of Civil Engineering (Aryan College of Engineering)

- **Technical Instruction:** Conducted technical orientation sessions and supervised student shake table dynamics competitions.